

5	(a) (i)	packet/discrete quantity/quantum (of energy) of e.m. radiation	B1	[1]
	(ii)	either $E = (6.63 \times 10^{-34} \times 3 \times 10^8)/(350 \times 10^{-9})$ or $E = (6.63 \times 10^{-34} \times 8.57 \times 10^{14})$ $E = 5.68 \times 10^{-19} \text{ J}$	M1 A0	[1]
	(iii)	0.5	B1	[1]
	(b) (i)	energy of photon to cause emission of electron <u>from surface</u> <i>either</i> with zero k.e <i>or</i> photon energy is minimum	M1 A1	[2]
	(ii)	correct conversion $\text{eV} \rightarrow \text{J}$ or $\text{J} \rightarrow \text{eV}$ seen once photon energy must be greater than work function 350 nm wavelength and potassium metal	B1 C1 A1	[3]
6	(a)	probability of decay	M1	